Md. Inzamamul Haque Shishir

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OBJECTIVE

Biomedical Engineering graduate from Jashore University of Science and Technology with three years of hands-on research in electrochemical biosensors. My work spans sensor fabrication and surface modification, nanomaterial synthesis, electrochemical characterization, and real-sample analysis, along with metabolomics-based biomarker research in breast cancer. Co-author of two peer-reviewed journal publications. I am passionate about building affordable, accessible diagnostic technologies, particularly for early cancer detection, through interdisciplinary research in biosensors, metabolomics, nanobiotechnology, wearable devices, and lab-on-chip systems. I am seeking graduate research opportunities where I can contribute to biomedical innovations with real-world impact.

EDUCATION

Bachelor of Science in Biomedical Engineering
Department of Biomedical Engineering
Jashore University of Science and Technology (JUST), Jashore-7408, Bangladesh
Graduated: July 2026 **CGPA: 3.20 / 4.00**

RESEARCH EXPERIENCE

Undergraduate Researcher (2023–2026), Laboratory of Nano-bio and Advanced Materials Engineering (NAME Lab)
Supervisor: Dr. Md. Zaved Hossain Khan
Responsibilities:

- Synthesized nanomaterials and nanocomposites for electrochemical biosensor applications.
- Performed electrochemical characterization
- Conducted real sample analysis using human blood serum.
- Assisted in experimental design, optimization, data interpretation, and scientific manuscript preparation.
- Collaborated with multidisciplinary researchers on biosensor development for clinical diagnostic applications.

RESEARCH & PUBLICATION

- Rayhan, R., **Shishir, M. I. H.**, Khaleque, M. A., Amin, M. R., Ali, M. R., Aly, M. A. S., Ayon, S. M., Saidur, R., Kim, T. H., Zaed, M. A., & Khan, M. Z. H. (2025). *A portable smartphone-based electrochemical sensing platform for rapid and sensitive detection of creatinine in blood serum.* **RSC Advances, 15**(30), 24917–24929. https://doi.org/10.1039/D5RA03128A
- Amin, M. R., Ali, M. R., Khaleque, M. A., Hossain, M. I., **Shishir, M. I. H**., Hasan, M. R., Aly, M. A. S., Saidur, R., Kim, T. H., Zaed, M. A., & Khan, M. Z. H. (2026). *Enhanced electrochemical sensing of L-cysteine in human blood serum using $Ti_3C_2T_x$/carbon black nanocomposite hybrid structure.* **Microchemical Journal, 221**,116817. https://doi.org/10.1016/j.microc.2026.116817

CONFERENCES

Participated in: (Category: Poster presentation)

- 23rd Science Council of Asia Conference (SCAC)
  Poster Titled: *A portable smartphone-based electrochemical sensor for rapid and sensitive detection of creatinine in blood serum.*
  Held on 30 November – 2 December 2024, Dhaka, Bangladesh
- International Biotechnology Conference 2025
  Poster Titled: *HealthTracker: A Machine Learning Enabled Dual Sensor Platform for Real Time Glucose and ECG Monitoring in Diabetic Care.*
  Held on 20 – 21 June 2025, BRAC University, Dhaka, Bangladesh

UNDERGRADUATE THESIS

"*Reproducible Metabolite Biomarkers in Breast Cancer: A Cross-Study Harmonization and Meta-Analysis***"**
Bachelor's Thesis | 2026 || Supervised by: Md. Faisal Mina, Lecturer
Department of Biomedical Engineering, Jashore University of Science and Technology, Jashore-7408, Bangladesh || Email: mf.mina@just.edu.bd

- Integrated three public LC-MS metabolomics datasets (**n = 337**) to identify reproducible plasma biomarkers for breast cancer.
- Developed a reproducibility-centered computational workflow using cross-study harmonization, FDR correction, effect-size filtering, and heterogeneity analysis.
- Identified five consistently depleted amino acid biomarkers and evaluated their diagnostic performance through pathway enrichment and machine learning.

Tools: Python • Metabolomics Workbench • HMDB • KEGG • Random Forest • Gradient Boosting • Statistical Meta-analysis

PROJECTS

(a) HealthTracker: Machine Learning–Enabled Dual Sensor Platform (2024)

- Designed a prototype healthcare monitoring platform integrating glucose sensing and ECG monitoring.
- Developed a real-time physiological monitoring workflow for diabetic patient care.

(b) Hospital Finder: AI-Based Emergency Healthcare Assistant (2026) Link

- Developed an AI-powered emergency healthcare assistant using Retrieval-Augmented Generation (RAG), Groq API, FastAPI, and PostgreSQL.
- Implemented intelligent hospital recommendations based on user location.
- Integrated personalized health data storage and first-aid guidance features.
- Designed the system to improve emergency healthcare accessibility.

LABORATORY & TECHNICAL SKILLS

Biosensing & Electrochemistry: Nanomaterial Synthesis, Electrode Surface Modification, Electrochemical Biosensor Fabrication, Real Sample Analysis
Characterization Techniques: FTIR, ATR
Molecular Biology Techniques: PCR, RT-PCR, Agarose Gel Electrophoresis
Programming: C**,** Python
Scientific Communication: Academic writing, paper publication, Scientific Manuscript Preparation, Experimental Design, Data Analysis

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Leadership Positions:

- General Secretary**, JUST Robo Society** (2024 - 2025)
  Jashore University of Science and Technology
- Secretary of Corporate and Public Relations, **JUST Research Society** (2024 - 2026)
  Jashore University of Science and Technology

Community Service:

- **Regular Blood Donor**

REFERENCES

(a) Dr. Md. Zaved Hossain Khan
Professor and Chairman
Department of Chemical Engineering
Director, NAME Lab (Nano-bio and Advanced Materials Engineering)
Jashore University of Science and Technology,
Jashore-7408, Bangladesh
Email: zaved.khan@yahoo.com

(b) Dr. Hassan Mohammad Al-Emran
Associate Professor and Chairman
Department of Biomedical Engineering
Jashore University of Science and Technology,
Jashore-7408, Bangladesh
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